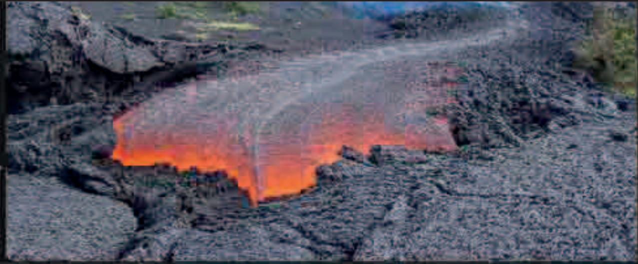
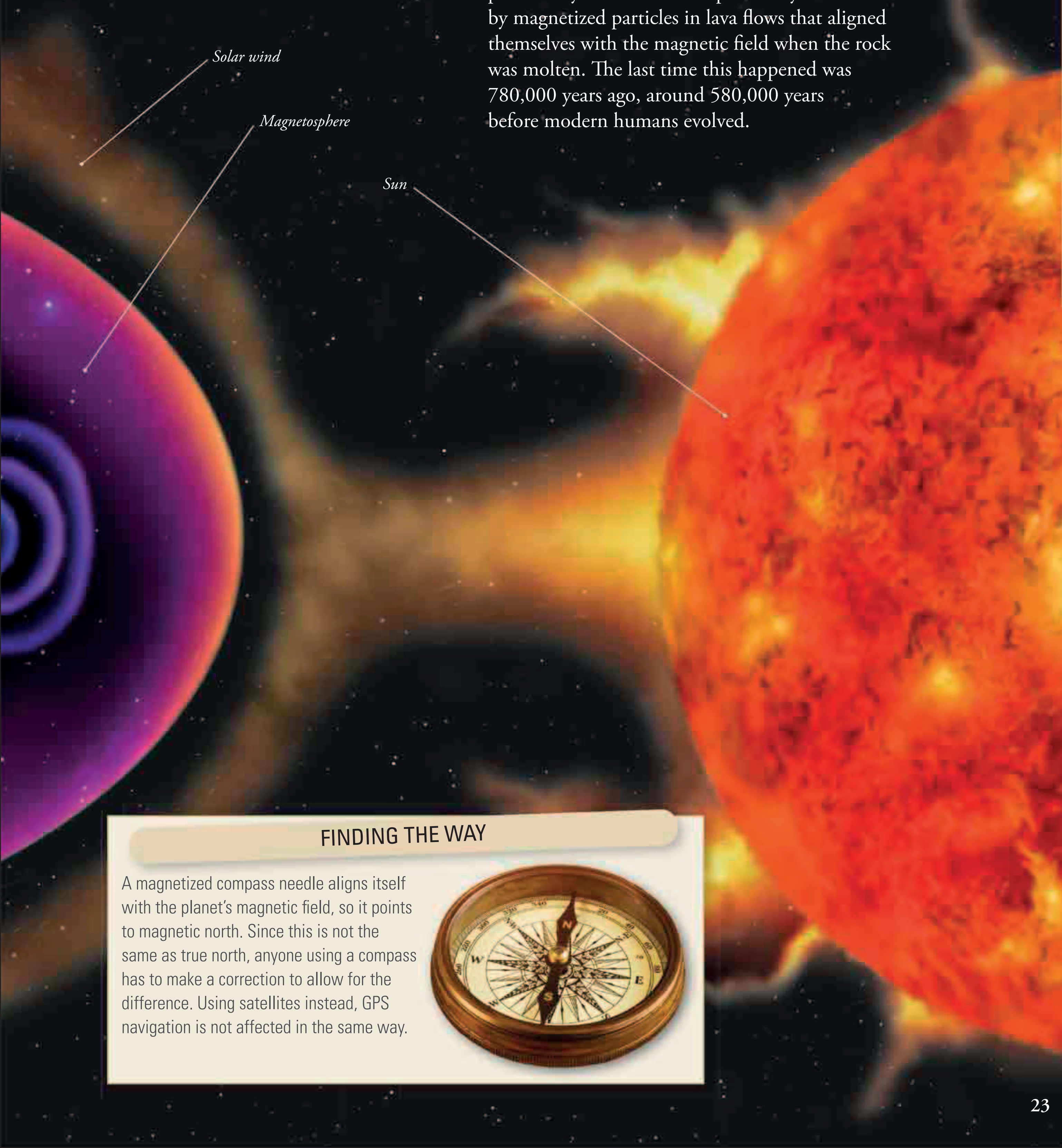


MAGNETIC SHIELD

The magnetosphere acts as a barrier to the solar wind. This stream of charged particles would otherwise strip away the atmosphere's ozone layer, which protects us from harmful ultraviolet radiation. The solar wind distorts the magnetosphere so it is compressed on the side facing the Sun, but extended on the other side.

MAGNETIC REVERSALS

Earth's magnetic field is so unstable that its poles may completely reverse. These reversals have taken place many times in Earth's past. They are recorded by magnetized particles in lava flows that aligned themselves with the magnetic field when the rock was molten. The last time this happened was 780,000 years ago, around 580,000 years before modern humans evolved.



FINDING THE WAY

A magnetized compass needle aligns itself with the planet's magnetic field, so it points to magnetic north. Since this is not the same as true north, anyone using a compass has to make a correction to allow for the difference. Using satellites instead, GPS navigation is not affected in the same way.